

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIC502 - MACHINE LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Machine Learning is a subset of:
A. Deep Learning B. Artificial Intelligence C. Neural Networks D. Robotics
 - 2 Which of the following uses labeled data?
A. Supervised Learning B. Unsupervised Learning
C. Reinforcement Learning D. None
 - 3 Which is an example of classification problem?
A. Predicting house prices B. Diagnosing a disease as positive/negative
C. Finding groups in data D. Reducing dimensions
 - 4 Overfitting occurs when:
A. Model performs well on test data B. Model generalizes well
C. Model learns noise in training data D. None of the above
 - 5 Which algorithm can be used for both classification and regression?
A. KNN B. Linear Regression C. Naive Bayes D. None
 - 6 What does the sigmoid function output range between?
A. -1 to 1 B. 0 to 1 C. $-\infty$ to ∞ D. 1 to ∞
 - 7 Which of these is sensitive to outliers?
A. K-Means B. Decision Tree
C. Linear Regression D. All of the above
 - 8 What does 'K' in KNN represent?
A. Number of classes B. Number of features
C. Number of neighbours D. Kernel used
 - 9 Naive Bayes assumes:
A. Feature independence B. Linear relationships
C. Non-linear decision boundaries D. Clustering nature
 - 10 In Random Forest, each tree is trained on:
A. Same data B. Different subset
C. Random noise D. Only features

- 11 Which function is used to train a model in scikit-learn?
 A. model.run() B. model.train()
 C. model.predict() D. model.fit()
- 12 Which of the following algorithm are not an example of ensemble learning algorithm?
 A.Random Forest B.Adaboost
 C.Gradient Boosting D.Decision Trees

Q2. Solve the following.

- A) What is Machine Learning? Differentiate between Traditional Programming and Machine Learning. 6
- B) Explain various categorization of Machine Learning in terms of required output. 6

Q3. Solve the following.

- A) What is Confusion Matrix? Explain Accuracy, Precision and Recall in detail. 6

B) 6

Actual Price	Predicted Price
250	265
110	140
500	480
200	215
330	290
490	515
670	750
210	210
435	420
375	285

From the above table calculate:

- a. MAE
 b. MSE
 c. RMSE

Q4. Solve Any Two of the following.

- A) How Regression is different from Classification? Explain Logistic regression in detail. 6

B) 6

Number of Crabs	Number of Clams Remaining
2	137
4	70
2	184
5	0
4	35
0	297
3	122
5	1
1	253
3	150

First, perform a linear regression on above dataset using the step-by-step instructions for calculating (slope) and (intercept) of the regression line. **What is the full equation for the regression line that you calculated?**

- C) Explain Gradient Descend in detail along with Learning Rate. 6

- Q5. Solve Any Two of the following.
- A) Explain the terms: A)Entropy B)Gini Index C) Decision Tree Overfitting 6
 - B) Explain Data Bagging and Feature selection in Random Forest. 6
 - C) What is Pruning? Explain different techniques of Pruning and Why Pruning is important. 6
- Q6. Solve Any Two of the following.
- A) Explain Naïve Bayes Classifier in detail along with advantages and disadvantages. 6
 - B) Explain KNN algorithm in detail. 6
 - C) Explain SVM along with its working in detail. 6

***** End *****